PERFORMANCE SPECIFICATION
DIGITAL TERRAIN ELEVATION DATA (DTED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the requirements within Defense Mapping Agency's (DMA) Digital Terrain Elevation Data Base which supports various weapon and training systems.

1.2 Purpose. The purpose of this specification is to assure uniformity of treatment among all mapping and charting elements engaged in a coordinated production and maintenance program for this product.

1.3 Classification. Digital Terrain Elevation Data (DTED) is produced at two different levels of detail. The two classes of DTED are known as DTED, Level 1; and DTED, Level 2.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the current Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, Defense Mapping Agency, ATTN: ATC, Stop A-13, 8613 Lee Highway, Fairfax, VA 22031-2137 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **AREA MCGT**

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-600001 - MC&G Accuracy Standard

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-9660 - DoD Produced CD-ROM Products

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

a. "Map Projections-A Working Manual", U.S. Geological survey Professional Paper 1395.

b. DIAM 65.18, Geopolitical Elements and Related Files.

c. DMAINST 8660.10, Release, Distribution, Handling, and Storage of DMA Digital Mapping, Charting, and Geodesy (MC&G) Data.

d. DMA C5231.1 MC&G Classification Management program.

e. DMAM 5200.1, DMA information Security Program.

f. DMATR 8350.2-B, Second Edition, 1 Sept. 91, DMA Technical Report, "Supplement to Department of Defense World Geodetic System 1984 Technical Report: Part II-Parameters Formulas and Graphics for the Practical Application of WGS 84".

g. DMA Procedural Instruction 813-102, Guidelines for Labeling The Defense Mapping Agency's CD-ROMs, and Printing and Finishing of Jewel Case Liners/Cardboard Sleeves, and Information Booklets.

h. DoD 5200.1-R, Information Security Program Regulation.

i. DoD 5220.22-M, Industrial Security for Safeguarding Classified Information.

(Copies of the above publication is available from the Defense Mapping Agency, Consumer Interface (OCI), 6001 MacArthur Boulevard, Bethesda, MD 201816-5001.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (See 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

a. ANSI X3.27-1978, American National Standard Magnetic Tape Labels for Information Interchange.

b. International Standard Organization (ISO) 9660: International Standard. Information Processing - Volume and File Structure of CD-ROM for Information Interchange. First Edition, 1988.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018).

c. Phillips/Sony Red Book Specifications.

d. Phillips/Sony Yellow Book Specifications.

(Application for copies should be addressed to the Sony Corporation of America, 5001 Forbes Boulevard, Lanham, MD, 20706).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Datum.

3.2.1 Horizontal datum. Horizontal Datum shall be the World Geodetic System (WGS 84). DTED shall be compiled to the current WGS datum or revised to be compatible with the current WGS datum.

3.2.2 Vertical datum. Vertical Datum shall be Mean Sea Level (MSL).

3.3 Accuracy.

3.3.1 Absolute accuracy.

3.3.1.1 Absolute horizontal accuracy.

90% Circular Error (C.E.) World Geodetic System (WGS) $\leq$ 50 meters

3.3.1.2 Absolute vertical accuracy.

90% Linear Error (L.E.) Mean Sea Level (MSL) $\leq$ 30 meters

3.3.2 Relative accuracy (point-to-point).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.3.2.1 Relative horizontal accuracy (point-to-point).

90% C.E. WGS ≤30 meters over a 1° cell

3.3.2.2 Relative vertical accuracy (point-to-point).

90% L.E. MSL ≤20 meters over a 1° cell

3.4 Security.

3.4.1 Security Classification. The security classification of the products generated by the use of these specifications will be the lowest practicable. When it is necessary to assign a security classification to the product, it shall be in accordance with established national security procedures.

This specification is unclassified, but its resultant products can be up to SECRET.

3.5 DMA Customer Help Desk.

For questions concerning this or other DMA Products or Services, please telephone the DMA Customer Help Desk, at 1-800-455-0899, Commercial 314-260-1236, or DSN 490-1236.

3.6 Product format.

a. A data file of DTED Level 1 or DTED Level 2 is a 1° by 1° cell defined by whole degree latitude and longitude lines on WGS. A DTED file shall not cross whole degree latitude or longitude lines.

b. In special cases, a partial DTED cell will be produced due to source constraints.

3.6.1 Adjacent data files. Adjacent data files shall not have gaps between them and the only overlap that exists is along adjacent boundaries. All adjacent boundaries shall be coincident.

3.7 Standard format. The DMA Standard Terrain Format is DMA's standardized system of recording terrain elevation data. The format is available on Compact Disc-Read Only Memory (CD-ROM), 8mm tape, and nine(9) track magnetic tape and intended for the production, storage and exchange of terrain elevation data. The additional characteristics of terrain elevation data on CD-ROM are described in 3.11.

3.7.1 Physical characteristics of magnetic tape.

3.7.1.1 Nine (9) Track Tape. The physical characteristics of Nine (9) track magnetic tape are as follows:

a. Length: 2400 feet (maximum).

b. Width: .5 inch.

c. Nine track recording format.

d. Odd parity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

e. Inter-record gap: .6 inch for 1600 bits per inch (BPI) Tapes; .3 inch for 6250 BPI Tapes.

f. Physical end-of-tape markers at the beginning (beginning-of-tape marker) and end of the tape (end-of-tape marker).

g. Density-recording methods.

(1) 1600 BPI/Phase encoded.

(2) 6250 BPI/Group Coded Recording (GCR). Preferred DMA data exchange format (shall not be used unless agreed to by sender and receiver).

3.7.1.2 Eight millimeter (8mm) Tape. The physical characteristics of Eight millimeter (8mm) magnetic tape cartridges are as follows:

a. Length: 112 meters (maximum).

b. Width: 8 mm.

c. Helical recording format.

d. Physical end-of-tape markers at the beginning (beginning-of-tape marker) and end of the tape (end-of-tape marker).

e. Density-recording methods. 2.3 GB or 5.0 GB per data cartridge.

3.7.2 Type of Input/Output (I/O). The individual records contain no system dependent generated control words.

3.8 Record characteristics on Magnetic Tape.

3.8.1 Recorded labels. American National Standard Magnetic Tape Labels for Information Interchange X3.27-1978. Recorded in ASCII code.

3.8.2 Data records.

a. Record size. Variable length, maximum 7214 bytes, minimum 14 bytes, modal (average) 2414 bytes.

b. Blocking factor. 1:1 (block size = record size).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

c. Record sequence.

```
VOL     1     (Volume Header Label)
HDR     1     (Header Label for file A)
UHL     1     (User Header Label for file A)
*
DSI           (Data Set Identification Record for file A)
ACC           (Accuracy Description Record for file A)
DATA          (for file A)
*
EOF     1     (End of File for file A)
UTL     1     (User Trailer Label for file A)
*
HDR     1     (Header Label for file B)
UHL     1     (User Header Label for file B)
*
DSI           (Data Set Identification Record for file B)
ACC           (Accuracy Description Record for file B)
DATA          (for file B)
*
EOF     1     (End of File for file B)
UTL     1     (User Trailer Label for file B)
*
*
```

Note: In the above sequence a Tape Mark (hardware end of file) is denoted by an "*".

3.9 File description. A data file of DTED is a cell defined by latitudes and longitudes of a geographic reference system. The terrain elevation information is expressed in meters. The locations of elevation posts are defined by the intersections of rows and columns within a matrix. The required matrix intervals, defined in terms of geographic arc seconds, vary according to latitude (see TABLE I for DTED Level 1, TABLE II for DTED Level 2).

Note: All values in seconds are in terms of arc measure.

TABLE I. Matrix intervals for DTED Level 1.

ZONE	LATITUDE	MATRIX INTERVAL			
		latitude		longitude	
I	0° - 50° North-South	3	x	3	seconds
II	50° - 70° North-South	3	x	6	seconds
III	70° - 75° North-South	3	x	9	seconds
IV	75° - 80° North-South	3	x	12	seconds
V	80° - 90° North-South	3	x	18	seconds

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

TABLE II. Matrix intervals for DTED Level 2.

ZONE	LATITUDE	MATRIX INTERVAL			
		latitude		longitude	
I	0° - 50° North-South	1	x	1	seconds
II	50° - 70° North-South	1	x	2	seconds
III	70° - 75° North-South	1	x	3	seconds
IV	75° - 80° North-South	1	x	4	seconds
V	80° - 90° North-South	1	x	6	seconds

3.10 Logical characteristics.

3.10.1 Data file structure. Arranged in 1° by 1° geographic areas. Each data file shall contain data falling within a single one degree cell. The reference origin for all data files shall be the southwest corner of the one degree cell. Multiple data files shall be arranged primarily by ascending latitude bands (90° South to 89° North), secondarily by ascending longitude (180° West to 179° East).

3.10.2 Files extent. To provide overlap between adjacent data files, the degree cell coverage in this standard includes the integer degree values on all sides of the area. Each data record has one point of overlap with the cell above and one with the cell below (if the record extends to the degree cell limits). Entire data records lying on integer degree longitude values shall exist also in the adjacent degree cell. Data files will not cross integer degree latitude or longitude lines. Adjacent data files shall not have gaps between them and the only overlap that exists is along adjacent boundaries. All data files derived from coincident boundaries of adjacent cells shall be comprised of duplicate data records.

3.10.3 Terrain elevation intervals. The latitude and longitude grid spacing will be in whole second intervals. Latitude spacing will always be consistent based upon the level of the data. Longitude spacing is dependent upon the level of the data and the geographic zone. (See TABLES I and II.)

3.10.4 Data value sequence. The elevations with a data record have a constant longitude value. The first data value is the southernmost known elevation and the last data value is the northernmost. No two data records shall have the same longitude value.

3.10.5 Data record sequence. Within a data file, the data records are arranged by ascending (west to east) longitude.

3.10.6 Data Block Checksum. The last four bytes of each data record contains a 32-bit checksum value. The checksum is computed algebraically using integer arithmetic by summing all header and elevation bytes contained in the record as 8-bit values. Each byte is considered an unsigned, 8-bit value for checksum calculation.

3.10.7 File structure diagram. (See TABLES III and IV.)

3.11 Data record characteristics.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

3.11.1 Numeric value. All elevation values are signed magnitude binary integers, right justified, 16 bits (2 bytes). The sign bit is in the high order position. Negative values are not complemented.

Note: The number of records is a function of the latitude and completeness of the cell coverage. A count of 1201 is for complete cells between latitudes 50°S and 49°N. Elevations are two-byte integers, high order first, and negatives are signed magnitude. Users may have to swap the bytes and/or convert negatives to the complement they use. This can be done by putting the low order byte first, then turning off bit 15 (the high order bit), then multiplying by -1. For positive numbers, only the bytes are switched.

3.11.2 Elevation values. The numeric value identified in 3.9.1 will allow a theoretical range of elevations between ± 32,767 meters; however in practice, the terrain elevation values shall not exceed + 9,000 meters or - 12,000 meters.

3.11.3 Partial cells. Partial cells may contain null values or may have missing elevations. Partial cells are not issued as standard DMA DTED.

3.11.3.1 Null elevations. Unknown elevations along a scan line which are bounded by known elevations values will contain the null value. The null value is represented by data values with all bits set to 1> bits within the structure of a data record. This null elevation value of - 32,767 meters is used as a place holder in the data record. Null elevation values are allowed in 1° cells (i.e., partial cells) which have not been fully compiled. Null elevation values shall not be contained in 1° cells (i.e., full cells) issued from DMA to users as a standard product.

3.11.3.2 Missing Elevations.

a. Elevation posts which are missing prior to the first known elevation are not represented. The location of the first known point is given by the latitude count field in the Data Record.

b. Elevation posts which are missing past the last known elevation are not represented. The last known elevation is immediately followed by a checksum.

c. Missing elevations are allowed in 1° cells (partial cells) which have not been fully compiled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

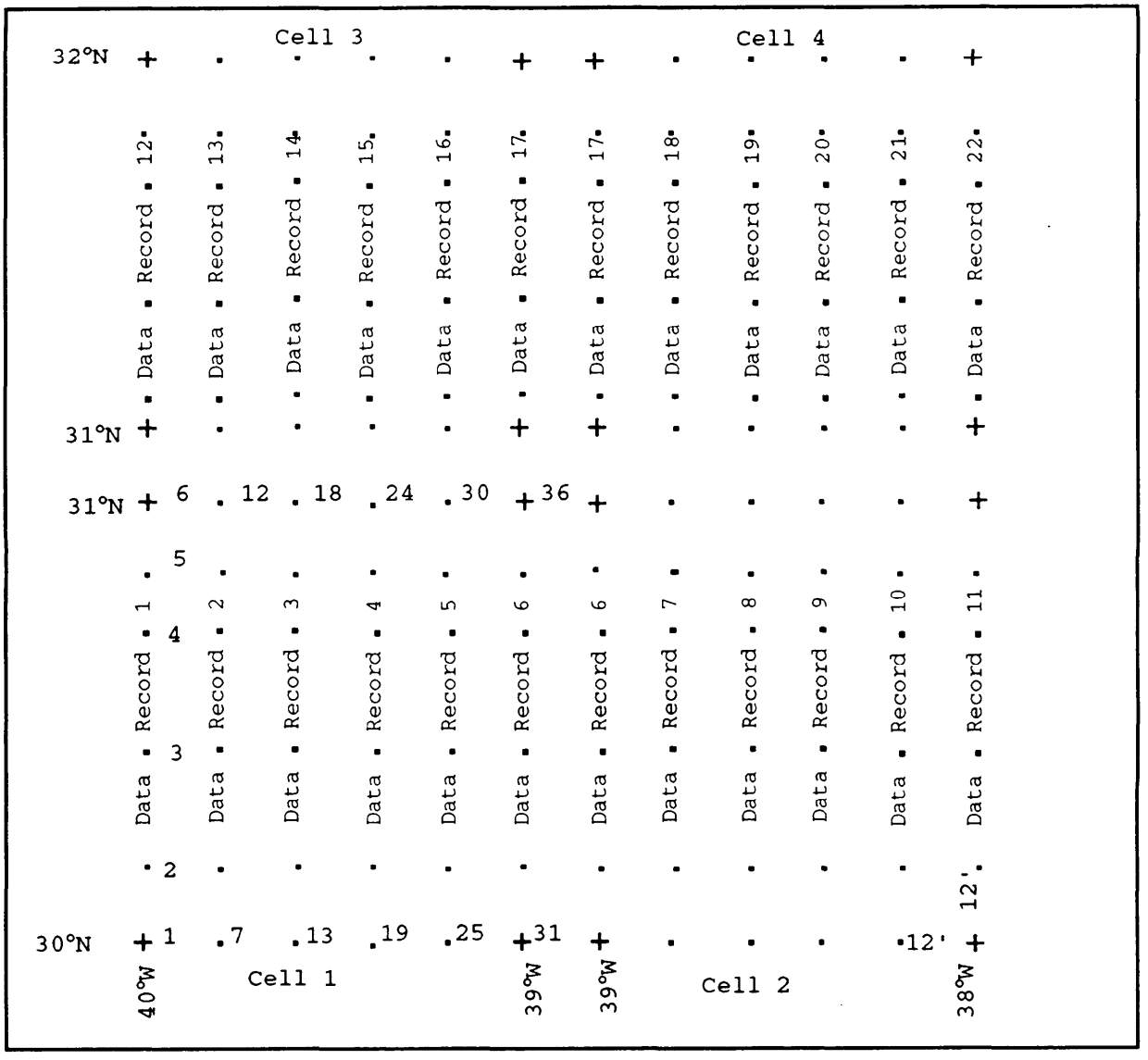
TABLE III. File Structure diagram-Terrain example.

FOUR 1° CELLS 12' LONGITUDE SPACING (NON STANDARD)			
DATA STRUCTURE SEQUENCE	DATA STRUCTURE TYPE	DATA STRUCTURE SEQUENCE	DATA STRUCTURE TYPE
1	VOL 1	32	HDR 1
2	HDR 1	33	UHL 1 31°N, 40°W
3	UHL 1 30°N, 40°W	34	*
4	*	35	DSI
5	DSI	36	ACC
6	ACC	37	Data Record 12
7	Data Record 1	38	Data Record 13
8	Data Record 2	39	Data Record 14
9	Data Record 3	40	Data Record 15
10	Data Record 4	41	Data Record 16
11	Data Record 5	42	Data Record 17
12	Data Record 6	43	*
13	*	44	EOF 1
14	EOF 1	45	UTL 1
15	UTL 1	46	*
16	*	47	HDR 1
17	HDR 1	48	UHL 1 31°N, 39°W
18	UHL 1 30°N, 39°W	49	*
19	*	50	DSI
20	DSI	51	ACC
21	ACC	52	Data Record 17
22	Data Record 6	53	Data Record 18
23	Data Record 7	54	Data Record 19
24	Data Record 8	55	Data Record 20
25	Data Record 9	56	Data Record 21
26	Data Record 10	57	Data Record 22
27	Data Record 11	58	*
28	*	59	EOF 1
29	EOF 1	60	UTL 1
30	UTL 1	61	*
31	*	62	*

Note: * = Tape Mark

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:54Z
Check the source to verify that this is the currentversion before use.

TABLE IV. File structure diagram-terrain example.



Notes: (1) In the above example (non-standard 12' latitude/longitude spacing), each data Record contains 6 elevations.

a. DTED Level 1 - A 3" standard 1° cell data record contains 1201 elevatons.

b. DTED Level 2 - A 1" standard 1° cell data record contains 3601 elevatons.

(2) Elevations along 1° boundaries are repeated for each 1° cell.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.12 Description of record contents. The following Standard Terrain Format is unique to DTED distributed on magnetic tape. For each of the following records, an ASCII character requires one byte (8 binary bits).

a. Volume Header Label (VOL).

Field Contents	Field Length In Characters	Character Start	Description
VOL	3	1	Recognition sentinel.
1	1	4	Fixed by standard.
	6	5	Reel Number. Six alpha-numeric characters identifying the physical reel. (Requires leading alpha and trailing numeric.)
Blank or Non blank	1	11	Non blank indicates restricted access, as the tape reel is privately owned.
Blanks	26	12	Unrequired available space.
Account Number	14	38	*Account number of owner of this tape reel. (DMA uses a maximum of 12 characters left-justified, space filled.)
Blanks	28	52	Fixed by standard.
1	1	80	Fixed by standard.

Note: *These fields, to be defined by the producer, may be left blank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

b. File Header Label (HDR).

Field Contents	Field Length In Characters	Character Start	Description
HDR	3	1	Recognition sentinel.
1	1	4	Fixed by standard.
Filename	17	5	*Left-justified filename The first 12 characters are referenced by the Executive System for comparison with the filename portion of the external filename.
UNIVAC	6	22	*Fixed as set identifier when referenced by system.
0001	4	28	*Reel sequence number within a file.
0001 - NNNN	4	32	*File sequence number within a reel.
0001 00	4 2	36 40	*Generation and version numbers which are fixed at 1 and 0.
bYYDDD	6	42	Creation date of tape. (A blank followed by two characters for the year, followed by three characters for the day (001 through 366) within the year date tape was written.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
ØYYDDD	6	48	*Expiration date of tape. Same format as creation date field. The date after which this tape reel may be considered as available for reallocation.
A space indicates unlimited access to this reel. 15_8 - This reel is catalogued (on tape). 35_8- This reel is catalogued with read key. 55_8 - This reel is catalogued with write key. 75_8 - This reel is catalogued with read and write key.	1	54	*Accessibility.
Block Count	6	55	*Fixed at zeros.
Qualifier	13	61	*Used by the Executive Operating System.(DMA uses a maximum of 12 characters left-justified space filled.)
Blanks	7	74	Fixed by standard.

Note: *These fields, to be defined by the producer, may be left blank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

c. User Header Label (UHL).

Field Contents	Field Length In Characters	Character Start	Description
UHL	3	1	Recognition sentinel.
1	1	4	Fixed by standard.
DDDMMSSH	8	5	Longitude of origin (lower left corner of data set; full degree value; leading zero(s) for all subfields: degrees, minutes and seconds). H is the Hemisphere of the data.
DDDMMSSH	8	13	Latitude of origin (lower left corner of data set; full degree value; leading zero(s) for all sub fields: (degrees, minutes and seconds). H is the Hemisphere of the data.
SSSS	4	21	Longitude data interval in seconds (decimal point is implied after third integer).
SSSS	4	25	Latitude data interval in seconds (decimal point is implied after third integer).
0000-9999 or Not Available (NA)	4	29	Absolute Vertical Accuracy in Meters. (With 90% assurance that the linear errors will not exceed this value relative to mean sea level (Right justified)).
T - Top Secret S - Secret C - Confidential U - Unclassified R - Restricted	3	33	Security Code. (Left justified)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
Unique Reference	12	36	*Unique reference number (Provides Number pointer to file containing detailed file description).
Number of longitude lines	4	48	Count of the number of longitude (profiles) lines
Number of latitude points	4	52	*Count of the number of latitude points per longitude line.
Multiple accuracy	1	56	0 - Single 1 - Multiple
Reserved	24	57	Unused portion for future use.

Note: *These fields, to be defined by producer, may be left blank.

d. Data Set Identification (DSI) Record.
Fixed Length = 648 ASCII Characters

Field Contents	Field Length In Characters	Character Start	Description
DSI	3	1	Recognition Sentinel.
T - Top Secret S - Secret C - Confidential U - Unclassified R - Restricted	1	4	Security Classification Code.
	2	5	Security Control and Release Markings. For DoD use only (DIAM 65-19).
	27	7	Security Handling Description. Other security description. (Free text or blank filled).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
	26	34	Reserved for future use. (Blank filled).
DTED1 or DTED2	5	60	DMA Series Designator for product level.
	15	65	Unique reference number. (For producing nations own use (free text or zero filled)).
	8	80	Reserved for future use. (Blank filled).
01-99	2	88	Data Edition Number.
A-Z	1	90	Match/Merge Version.
YYMM	4	91	Maintenance Date. (Zero filled until used.)
YYMM	4	95	Match/Merge Date. (Zero filled until used.)
0000 or ANNN	4	99	Maintenance Description Code. (Zero filled until used.)
CCAAABBB	8	103	Producer Code.(Country - Free Text) (DIA Country Codes used for first 2 characters).
	16	111	Reserved for future use. (Blank filled).
AAAAAAAAA	9	127	Product Specification. (Alphanumeric field)
00-99	2	136	Product Specification Amendment Number.
YYMM	4	138	Date of Product Specification.
MSL	3	142	Vertical Datum (Mean Sea Level).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
WGS84	5	145	Horizontal Datum Code (Current Version World Geodetic System).
	10	150	Digitizing/Collection System. (Free text).
YYMM	4	160	Compilation Date. (Most descriptive year/month).
	22	164	Reserved for future use. (Blank filled).
DDMMSS.SH	9	186	Latitude of origin of data– leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSS.SH	10	195	Longitude of origin of data– leading zero for values less than 100; H is the hemisphere of the data.
DDMMSSH	7	205	Latitude of SW corner of data, bounding rectangle– leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSSH	8	212	Longitude of SW corner of data, bounding rectangle– leading zero for values less than 100; H is the hemisphere of the data.
DDMMSSH	7	220	Latitude of NW corner of data, bounding rectangle– leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSSH	8	227	Longitude of NW corner of data, bounding rectangle– leading zero for values less than 100; H is the hemisphere of the data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
DDMMSSH	7	235	Latitude of NE corner of data, bounding rectangle—leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSSH	8	242	Longitude of NE corner of data, bounding rectangle—leading zero for values less than 100; H is the hemisphere of the data.
DDMMSSH	7	250	Latitude of SE corner of data, bounding rectangle—leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSSH	8	257	Longitude of SE corner of data, bounding rectangle—leading zero for values less than 100; H is the hemisphere of the data.
DDDMMSS.S	9	265	Clockwise orientation angle of data with respect to true North. (Will usually be all zeros for DTED.)
SSSS	4	274	Latitude interval in tenths of seconds between rows of elevation values. (Decimal point is implied after third integer.)
SSSS	4	278	Longitude interval in tenths of seconds between columns of elevation values. (Decimal point is implied after third integer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
0000-9999	4	282	Number of Latitude lines, Actual count - Number of latitude points (rows that contain data).
0000-9999	4	286	Number of Longitude lines, Actual count - Number of longitude points (columns that contain data).
00 or 01-99	2	290	Partial Cell Indicator 00 = Complete 1° cell 01-99 = % of coverage completed.
	101	292	Reserved for DMA use only. (Blank filled.)
	100	393	Reserved for producing nation use only. (Free text or blank filled.)
	156	493	Reserved for future use. (Blank filled.)

e. Accuracy Description (ACC) Record.
Fixed Length = 2700 ASCII Characters

Field Contents	Field Length In Characters	Character Start	Description
ACC	3	1	Recognition Sentinel.
0000-9999 or Not Available (NA)	4	4	*Absolute Horizontal Accuracy of Product in meters
0000-9999 or Not Available (NA)	4	8	*Absolute Vertical Accuracy of Product in meters
0000-9999 or Not Available (NA)	4	12	*Relative (Point-to-Point) Horizontal Accuracy of Product in meters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
0000-9999 or Not Available (NA)	4	16	*Relative (Point-to-Point) Vertical Accuracy of Product in meters.
	4	20	Reserved for future use. (Blank filled.)
	1	24	Reserved for DMA use only.
	31	25	Reserved for future use. (Blank filled.)
00 or 02-09	2	56	Multiple Accuracy Outline Flag. 00 = No accuracy subregions provided. 02-09 = Number of accuracy subregions per 1° cell (maximum 9).

Note: *If Product has sub regional accuracies, the overall accuracy of the product will be the worst accuracy.

Start of Accuracy Sub region Description. Repeat to maximum of nine times. Blank fill all unused accuracy sub regions or unused coordinate pairs within a sub region. (1 Sub region = 284 ASCII Characters). Refer to 3.10.5.1 for accuracy sub region description.

0000-9999 or Not Available (NA)	4	58	Absolute Horizontal Accuracy of Sub region in meters
0000-9999 or Not Available (NA)	4	62	Absolute Vertical Accuracy of Sub region in meters.
0000-9999 or Not Available (NA)	4	66	Relative (Point-to-Point) Horizontal Accuracy of Sub region in meters.
0000-9999 or Not Available (NA) region	4	70	Relative (Point-to-Point) Vertical Accuracy of Sub in meters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Characters	Character Start	Description
03-14	2	74	Number of coordinates in accuracy sub region outline. (Maximum of 14 coordinate pairs. Coordinates are input clockwise. Implied closing from last to first coordinate pairs.)
	Start of Coordinate Pair Description. Repeat to maximum of fourteen times to outline sub region. Blank fill all unused accuracy sub regions and unused portions of sub regions.		
DDMMSS.SH	9		Latitude–leading zero for values less than 10; H is the hemisphere of the data.
DDDMMSS.SH	10		Longitude–leading zero for values less than 100; H is the hemisphere of the data.
	End Coordinate Pair Description		
	End Accuracy Sub region Description		
	18	2614	Reserved for DMA use only.
	69	2632	Reserved for future use.

f. Data Record Description. Each elevation is a true value referenced to mean sea level datum recorded to the nearest meter. The horizontal position is referenced to precise longitude-latitude locations in terms of the current World Geodetic System (WGS), determined for each file by reference to the origin at the southwest corner. The elevations are evenly spaced in latitude and longitude at the interval designated in the user header label in South to North profile sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Field Contents	Field Length In Bytes	Description
252_8	1	Recognition Sentinel.
Data block count	3	Sequential count of the block within the file, starting with zero for the first block (Fixed Binary).
Longitude count	2	Count of the meridian. True longitude = longitude count x data interval + origin (Offset from the SW corner longitude) (Fixed Binary).
Latitude count	2	Count of the parallel. True latitude = latitude count x data interval + origin (Offset from the SW corner latitude) (Fixed Binary).
Elevation 1	2	True elevation value of point 1 of meridian in meters (Fixed Binary).
Elevation 2	2	True elevation value of point 2 of meridian in meters (Fixed Binary).
Elevation N	2	True elevation value of point N of meridian in meters (Fixed Binary).
Checksum	4	Algebraic addition of contents of block. Sum is computed as an integer summation of 8-bit values (Fixed Binary).

Note: Fixed Binary denotes signed magnitude, right-justified binary integers.

g. File Trailer Label (EOF).

Field Contents	Field Length In Characters	Description
EOF	3	Recognition Sentinel.
1	1	Fixed by standard.

Note: (See HDR for remainder of EOF fields.)

h. User Trailer Label (UTL)

Field Contents	Field Length In Characters	Description
UTL	3	Recognition Sentinel.
1	1	Fixed by standard.

Note: (See UHL for remainder of UTL fields.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.13 Explanation of records and fields (DTED).

3.13.1 Volume header label (VOL). This record is required for labeled tapes in accordance with American National Standards Institute (ANSI) standard X3.27-1978, Magnetic Tape Labels for Information Interchange.

3.13.2 File header label (HDR). This record is required for labeled tapes in accordance with ANSI standard X3.27-1978, Magnetic Tape Labels for Information Interchange.

3.13.3 User Header Label (UHL). ANSI standard allows an optional user header label in the first file of a labeled tape. Several computer manufacturers have implemented tape labeling in such a way that the user header label in the first file of the tape is inaccessible. This record is maintained for minimum impact to users not desiring to use the DSI record, but all information in it is in the DSI record as well.

3.13.3.1 Fields(UHL). The following are explanations of fields within the UHL.

a. Longitude of Origin - Origin is always a full degree value even though the format allows values to be expressed to the second.

b. Latitude of Origin - Origin is always a full degree value even though the format allows values to be expressed to the second.

c. Seconds Longitude Interval - A cell of DTED is North-South oriented with columns of elevation posts running from south to north. The longitude interval is the East-West distance between the columns expressed as tenths of seconds.

d. Seconds Latitude Interval - The spacing between the elevation posts within a column (i.e., the distance between the rows) is the latitude interval expressed in tenths of a second.

3.13.4 Data Set Identification (DSI) Record. This record provides identification and security information relating to the DTED. The record is fixed length consisting of 648 ASCII characters. Each character is represented by 1 tape byte or 8 bits. Certain fields in the DSI record are duplicated in the UHL/UTL. These fields are required to match. Users may process the data using only the information in the DSI record if desired.

Note: Numeric values shall be right justified with leading zeros. Alpha and alphanumeric characters shall be left justified unless 'free text' is indicated in description.

3.13.4.1 Fields(DSI).

a. Security control and release markings - This field may contain the two character codes from DIAM 65-19 or may be blank filled

b. DMA series designator - Five character code to uniquely identify a product. DTED1 for Level 1, or DTED2 for Level 2.

c. Unique reference number - Free text or zero filled.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

d. Data edition number - The number assigned to the data indicating either original compilation (Edition 1) or subsequent replacements of the data (Editions 2, 3, etc.) in the Data Base (DB) to achieve accuracy requirements (recompilation) or currency/specification requirements (revision).The data edition number does not reflect the number of replacements made to the data to effect boundary matches.

e. Match/merge version - The number of times an edition of the data was changed to effect boundary continuity with adjacent data in the DB. Alphabetic Code A denotes original release of the edition, B - Z designates data change for boundary continuity.

f. Maintenance date - YYMM (year and month); The date existing DB data was either revised (updated) to meet the currency requirements (or to effect specification changes), or recompiled to meet accuracy requirements. When the existing data is only revised (horizontal position or vertical values are not significantly changed) the maintenance date will reflect the date of the revision, but the compilation date will not be changed. It will continue to reflect the date of the original compilation. However, when the data is subjected to a major recompilation, the Compilation Date and the Maintenance Date will both be changed to reflect the date of the recompilation.

g. Match/merge date - YYMM (year and month); The latest date the data in the database (DB) was changed to effect continuity with adjacent data. This data corresponds to the Match/Merge Version Code.

h. Maintenance description code - Zero filled until used.

i. Producer code - The first two characters (left justified) indicate the producing nation and are from DIAM 65-18 - Geopolitical Elements and Related Files. The last six characters are to be used at the discretion of the producer. Blanks are acceptable. Example codes:

Belgium	BE	Netherlands	NL
France	FR	Norway	NO
United Germany	GM	United Kingdom	UK
Italy	IT	United States	US

j. Product specification - Identifies the product specification containing the compilation and accuracy requirements used to produce the data. Data produced to this specification will contain MIL-DTL-8902A.

k. Product specification amendment and change number - Indicates the highest numbered amendment or change used to produce the data. The first character identifies the Amendment number and the second character identifies the Change Notice number. (Amendment 0, Change 1-01; Amendment 2, Change 2-22; etc.)

l. Date of product specification - Identifies the published date of the product specification used to produce the data. Data produced to this specification will contain 9505.

m. Vertical datum code - Currently MSL.

n. Horizontal datum code - Identifies the version of WGS to which the product is horizontally controlled. During original compilation DTED uses the WGS datum then in effect for DMA use. Upon transition to a new WGS datum the existing DTED can be:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

(1) Recompiled from an original source; and therefore carry a new Edition number.

(2) Recompiled by mathematically adjusting the elevation values; and therefore carry a new Edition number.

(3) Revised by defining the same elevations as compatible with the new WGS datum within the accuracy evaluations quoted; and therefore carry the same Edition number but with a new Maintenance Date.

o. Digitizing collection system - Identifies the equipment used to collect the elevation values from the source material used, e.g., AGDS, LIS, UNAMACE, AS11, DE/S, PG/S, etc.

p. Compilation date - YYMM (year and month); The date the data was either originally compiled (Edition 1) or the date existing data was subjected to a major recompilation which involved significant changes to the horizontal positions and vertical values. (Edition 2, 3, 4, etc.)

q. Latitude of origin - Expressed in degrees, minutes, seconds and tenths of seconds with N or S to indicate hemisphere (always a full degree value even though the format allows value to be expressed to tenths of seconds).

r. Longitude of origin - Expressed in degrees, minutes, seconds and tenths of seconds with E or W to indicate hemisphere (always a full degree value even though the format allows value to be expressed to tenths of seconds).

3.13.5 Accuracy description record (ACC). The accuracy record provides accuracy information relating to the DTED. The record is fixed length consisting of 2700 ASCII characters. Each character is represented by 1 byte or 8 bits. The record allows for up to nine accuracy sub regions within the product should the accuracies of various portions of this product differ. Each sub region may consist of up to fourteen coordinate pairs to accurately portray the outline. Coordinates are input clockwise. Unused coordinate pairs are blank filled.

Note: Numeric values shall be right justified with leading zeros. Alpha and alphanumeric characters shall be left justified.

3.13.5.1 Accuracy sub region description. A sub region description contains the sub region accuracies, the number of sub region coordinate pairs and the actual coordinate pairs which define the sub region area. Unused sub regions and/or unused coordinate pairs within a sub region will be blank filled. The sub regions will be organized as follows:

a. The first coordinate pair of the first sub region must be the southwest corner of the bounding rectangle.

b. The first coordinate pair of each sub region must be the most southern and western point in the sub region.

c. The first coordinate pair of each sequential sub region must be arranged in a clockwise direction starting with the first sub region. Sub regions which contain at least part of a degree line will take precedence and be numbered clockwise, starting in the SW corner and proceeding around the

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:54Z
Check the source to verify that this is the current version before use.

four bounding degree lines. Remaining sub regions will be numbered in a south to north, west to east manner based on initial point within each sub region.

d. Coordinate pairs within a sub region must be arranged in a clockwise direction.

e. Each sub region coordinate pair must be located on or within the bounding rectangle.

f. Each coordinate pair in the bounding rectangle must be represented as a sub region coordinate pair.

g. Sub region areas must not overlap or leave gaps. The entire area defined by the bounding rectangle must be covered by sub region areas (i.e. the sum of the sub regions must equal the 1° product area).

h. The first and last coordinate pairs of a sub region shall not be the same. Closure of the sub region area is implied.

i. A sub region can only be generalized if all of the sub region descriptions have been used.

j. Sub region format:

Sub region Number	Field Length in Characters	Character Start
1	284	58
2	284	342
3	284	626
4	284	910
5	284	1194
6	284	1478
7	284	1762
8	284	2046
9	284	2330

3.14 DTED on Compact Disc-Read Only Memory (CD-ROM).

3.14.1 General.

3.14.1.1 CD-ROM content. The content of the DTED CD-ROM product includes, in addition to the DTED files, a Digital Mean Elevation Data (DMED) file, and zero or more Gazetteer files. The DMED file provides statistics about the elevations in each 15' x 15' area of each DTED cell. The Gazetteers provide the geographic coordinates for places, cities and other important landmark features. In addition, there are four text files that provide helpful information pertaining to the data on the DTED CD-ROM. Additional information concerning CD-ROMs can be found in the DoD Military Handbook titled, MIL-HDBK-9660 DoD Produced CD ROM Products.

3.14.1.2 CD-ROM characteristics. The physical characteristics of the compact disc media are specified by the Philips/Sony Red Book. The digital data recording characteristics of the CD-ROM are specified by the Philips/Sony Yellow book. Logical formatting, file directory structure, and labeling of MC&G data on CD-ROM conforms to International Standard Organization (ISO) 9660, level 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.14.2 Directories. Each DTED CD-ROM contains several hundred DTED cells, a gazetteer, a DMED file, a text file describing the disc contents including a user's guide, and a text file describing an Operational Navigation Chart (ONC) directory to access the DTED. A series of directories point to the above listed files resident on the CD-ROM. At the highest level, there are three directories (DTED, GAZETTE, and TEXT) and one file (DMED). Only the DTED directory contains sub directories. Within the DTED directory, there is a sub directory for each longitude containing cells. The sub directory name format is: <E/W>DDD, where DDD is degrees. For example, the sub directory name for DTED files contained between the meridians 127E and 128E would be : E127. The DTED cell file-name format is: <N/S>DD.DT1, where DD is degrees. For example, the file-name format for a DTED cell whose southwest corner is 38N and 127E would be: N38.DT1, or starting from the root directory, \DTED\E127\N38.DT1. FIGURE 1 is a Warnier diagram depicting the logical file structure (page 35).

3.14.3 CD ROM labeling and packaging.. CD ROM labeling, and labeling on the cardboard sleeve, or jewel case liner/information booklet, as applicable, shall be in accordance with DMA PI 813-102, Guidelines for Labeling The Defense Mapping Agency's CD-ROMs, and Printing and Finishing of Jewel Case Liners/ Cardboard Sleeves, and Information Booklets. Method of packaging (cardboard sleeves or jewel case) shall be specified in the contract (see 5.1).

3.14.3.1 CD labeling. Labeling of DTED CDs shall be in accordance with standard elements identified in the reference figures in DMA PI-813-102.

3.14.4 DTED files. A DTED Level 1 cell is a single file containing the following information subset of the standard DMA DTED tape distribution format. The logical tape format/label records VOL1, HDR1, EOF1, UTL1, and hardware end of file marks are not included in the file, since the CD-ROM is logically formatted/labeled according to ISO 9660.

a.	User Header Label (UHL: 80 bytes)	1
b.	Data Set Identification Record (DSI: 648 bytes)	81
c.	Accuracy Record (ACC: 2700 bytes)*	729
d.	Data Records (1201 records at 2414 bytes/record)**	3429,5843, 8257, etc.

* To meet current distribution security requirements of producing only unclassified CD DTED, when there are multiple accuracy sub regions they are left blank (bytes 58-2613 of the ACC record) and the multiple accuracy outline flag is set to 10 (bytes 56-57 of the ACC record).

** The number of records is a function of the latitude and completeness of the cell coverage. A count of 1201 is for complete cells between latitudes 50°S and 49°N. Elevations are two-byte integers, high order first, and negatives are signed magnitude. Users may have to swap the bytes and/or convert negatives to the complement they use. This can be done by putting the low order byte first, then turning off bit 15 (the high order bit), then multiplying by -1. For positive numbers, only the bytes are switched.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

3.14.5 Digital Mean Elevation Data (DMED).

3.14.5.1 DMED file. The Digital Mean Elevation Data (DMED) file contains, for each 15' x 15' area of a 1°x1° cell, the minimum and maximum elevation, the mean elevation, and the standard deviation. The DMED file is a series of 394-byte records. The first record is a header giving the extremes of the minimum bounding rectangle (MBR) (in degrees) encompassing the cells on the CD. For example, the value: N30N36E020E032 followed by 380 spaces indicates that the MBR is 6° high and 12° wide. The next record has the data for the extreme southwest 1° cell in the MBR. The third record is for the 1° cell above that, and so forth, to the top of the MBR. Moving eastwardly, and until the edge of the MBR is reached, the next column of 1°x1° cells is added to the file with placement being from south to north. If a cell in the MBR is not included on the CD, its DMED record consists of its coordinates followed by 387 spaces (ASCII blanks).

3.14.5.2 Individual cell record. An individual cell record contains the geographic coordinates of the cell's southwest corner, the data edition number, the match/merge version of the cell from which the DMED is calculated and the following information about each 15' x 15' area of the cell: the minimum and maximum elevations, the mean elevation, and the standard deviation of the elevations. Data are all upper-case ASCII characters. Elevation values are to the nearest meter. In a 1° cell, the elevations in the three rows and three columns which divide the 15' x 15' areas from each other are counted in two areas. The order in which the 16 areas of a cell follow the header is:

4	8	12	16
3	7	11	15
2	6	10	14
1	5	9	13

Field Contents	Field Length in Characters	Start	Character Description
H	1	1	Hemisphere (N or S)
DD	2	2	Latitude of SW corner of cell
H	1	4	Hemisphere (E or W)
DDD	3	5	Longitude of SW corner of cell
01-99	2	8	Data edition number
A-Z	1	10	Match/merge version
integer	6	11	Minimum elevation, in meters, of SW area. If negative, sign will be the next place left of most significant digit
integer	6	17	Maximum elevation of SW area
integer	6	23	Mean elevation of SW area
space	1	29	Not used
integer	5	30	Standard deviation about mean for SW area integer
	6	35	Minimum elevation of area 2
	41		Continue as for area 1
	24	59	Repeat for area 3
	24	83	Repeat for areas 4-16

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

3.15.6 GAZZETTER.

3.15.6.1 GAZETTE directory. The gazette directory has files GAZETTE.KEY and GAZETTE.DIR, plus two files for each gazette (country) with the name convention <country>.GAZ and <country>.HSH. The country name may be abbreviated. If the minimum bounding rectangle of a CD touches any country, the entire gazetteer for that country is included, resulting possibly in coordinates extraneous to the geographic coverage.

3.15.6.2 Primary file. The primary file is <country>.GAZ. It consists of variable-length ASCII alpha-numeric records, one entry for each name. Alphabetic characters are upper case. Each record consists of a designation code in bytes 1-4, the latitude in bytes 5-11 (DDMMSSH), the longitude in bytes 12-19 (DDDMMSSH), a space in byte 20, and a variable length name field starting in byte 21. The name field contains a name, or a name followed by "SEE" followed by another name. The name field is followed by a carriage return (CR), a line feed (LF), then the next record. Even though most coordinates are rounded to the nearest minute, seconds are included in the format to accommodate future data. The seconds contain the digits 00. Entries are ordered by the ASCII collating sequence of the names.

Field Contents	Field Length in Characters	Character Start	Description
alpha	4	1	Designation code
integer	2	5	Degrees lat (00-90)
integer	2	7	Minutes lat (00-59)
integer	2	9	Seconds lat (00-59)
alpha	1	11	Hemisphere (N or S)
integer	3	12	Degrees lon (000-180)
integer	2	15	Minutes lon (00-59)
integer	2	17	Seconds lon (00-59)
alpha	1	19	Hemisphere (E or W)
alpha	1	20	Space
alpha	var	21	Name field
ASCII	2	var	CR LF, hex 0D 0A

3.15.6.3 Hash file. The hash, or index, file is <country>.HSH. The hash file is a convenience for the user who wants a quick pointer into the primary file. It consists of four-byte records which are 32-bit integers (high byte first). It is based on the first three or four characters of the names in the primary file. Three characters are used for gazetteers with less than 50,000 entries, and four are used for larger ones. The binary integer in bytes 1-4 of the hash file tells how many characters were used. Characters considered for the computation are given in the following table along with the hexadecimal ASCII value and the decimal hash value. All other characters take the value of the space.

ASCII Character	Hex	Hash
space	20	0
'	27	1
,	2C	2
-	2D	3
.	2E	4
/	2F	5
A-Z	41-5A	6-31
grave	60	32
tilde ~	7E	33

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

3.15.6.4 GAZETTE.KEY. The key file is a dictionary of the 4-letter designation codes. It is a text file of variable length records with designation code in bytes 1-4, space in 5, variable length explanation, and CR LF.

3.15.6.5 GAZETTE.DIR. This text file gives the abbreviated country names, and expands them to their full length.

3.15.7 Textual information.

3.15.7.1 READ.ME. An ASCII text file named READ.ME contains a description of the disc contents including the geographic limits of the compact disc, a count of the cells, a location diagram (text array) showing which cells in the minimum bounding rectangle are resident on the disc, and a user's guide.

3.15.7.2 ONC.DIR. An ASCII file named ONC.DIR provides, via ONC tiles, a directory to the DTED cells on the CD-ROM.

3.15.8 User's responsibilities.

a. Procedures applicable for protection of classified data are contained in DMA Manuals: 5200.1, DMA Information Security Program; C5231.1, MC&G Classification Management Program; and DoD 5220.22-M, Industrial Security for Safeguarding Classified Information.

b. The DTED on CD-ROM or magnetic tape is MC&G Property for purposes of the Defense Federal Acquisition Regulations (DFARS). Digital products produced by DMA in support of DoD requirements are Defense Information. While classification may not be required, the increasing size of the completed data bases represents a national resource. Therefore, DMA Instruction 8660.10, Release, Distribution, Handling, and Storage of Defense Mapping Agency (DMA) Digital Mapping, Charting, and Geodesy (MC&G) Data, establishes policies and procedures limiting unconstrained access. Recipients of DMA digital data must protect their holdings against misuse or loss. Whenever the digital products are not under supervision of authorized persons, the data must be secured in a container or vault that provides physical evidence of unauthorized tampering or forced entry.

c. Reproduction for further distribution outside the primary recipient's organizational structure of any amount of DTED is not authorized without the written permission of the Defense Mapping Agency. Requests for such permission are to be sent to:

Director, Defense Mapping Agency
Attn: ATIAD (Release Office)
8613 Lee Highway
Fairfax, VA 22031-2137

d. Digital data derived from data produced by other nations under cooperative agreements may be subject to additional restrictions, copyrights, or classification. These restrictions must be honored when such digital data is distributed.

e. The data structure and information content of DMA's digital products are prescribed by DMA Product Specifications or Military Specifications. Deletions, additions, or modifications to DMA's digital files

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

by users become local transaction files (no longer DMA products) and are the responsibility of the user to control.

f. Destruction notice for DTED CD-ROMs: Unclassified DTED CD-ROMs: Destroy by any method that will prevent disclosure of contents or reconstruction of the disc. Classified DTED CD-ROMs: Follow the procedures in DoD 5220.22-M, Industrial Security Manual, Section 11-19 or DoD 5200.1-R, Information Security Program Regulation, Chapter IX.

g. Correspondence concerning the quality of the CD-ROM shall be forwarded to:

Director, Defense Mapping Agency- Bethesda
Attn: ODGI
4600 Sangamore Road
Bethesda, Maryland 20816-5003

3.16 Copyright.

a. Copyright protection is asserted for all products generated by these specifications which are distributed outside of the United States. No domestic copyright will be asserted.

b. The copyright notice (with year of production inserted)states:

© COPYRIGHT (year of production) BY THE UNITED STATES GOVERNMENT.
NO COPYRIGHT CLAIMED UNDER TITLE 17 U.S.C.

3.17 CD-ROM labeling. See DMA Procedural Instruction PI 813-102 for guidelines for labeling Defense Mapping Agency CD-ROMs, and printing and finishing of jewel case liners/cardboard sleeves, and information booklets.

3.17.1 Information booklet. An information booklet will provide general information (similar to the READ.ME file) about the contents of the CD-ROM, handling instructions, distribution/user information, and DMA points of contact.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article inspection is required (see 3.1), it shall be examined as specified in 4.3.1, and tested as specified in 4.3.2.

4.3 Conformance inspection. Quality conformance inspection shall include the examination of 4.3.1 and the tests of 4.3.2.

4.3.1 Examination. The database shall be examined for compliance with the requirements specified in section 3. Unless a waiver has been granted non compliance with any of the specified requirements shall constitute cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

4.3.2 Tests. A sample determined by the contracting officer shall be read back after generation to ensure that the requirements specified in section 3 have been met.

4.4 Government furnished material. The contractor shall not duplicate, copy, or otherwise reproduce the MC&G property for purposes other than those necessary for performance of the contract.

4.5 Government property surplus. At the completion of performance of the contract, the contractor, as directed by the contracting officer, shall either destroy or return to the Government all government-furnished MC&G property not consumed in the performance of the contract.

5. PACKAGING

5.1 Packaging. For acquisitions purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's Systems Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. This specification defines the content of Digital Terrain Elevation Data (DTED) which supports various weapon and training systems.

6.2 Acquisition Requirements. When this specification is used in acquisition, the applicable issue of the Department of Defense Index of Specifications and Standards (DODISS) must be cited in the solicitation (See 2.1.1 and 2.2).

6.3 Supersession. This specification supersedes MIL-D-89020 Military Specification for Digital Terrain Elevation Data, 28 May 1993.

6.4 Definitions.

6.4.1 Accuracy. The degree of conformity with which horizontal position and vertical values are represented on a map, chart, or related product in relation to an established standard.

a. Horizontal accuracy, absolute. The uncertainty in the horizontal position of a point with respect to the World Geodetic System caused by random and uncorrected systematic errors. The value is expressed as a circular error at the 90% confidence level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

b. Horizontal accuracy, relative or point-to-point. The uncertainty in horizontal position between two points caused by random errors. The value is expressed as a circular error at the 90% confidence level.

c. Vertical accuracy, absolute. The uncertainty in the height of a point with respect to Mean Sea Level caused by random and uncorrected systematic errors. The value is expressed as a linear error at the 90% confidence level.

d. Vertical accuracy, relative or point-to-point. The uncertainty in height between two points caused by random errors. The value is expressed as a linear error at the 90% confidence level.

e. Random vertical accuracy. The uncertainty in the height of a point caused by random error only. The value is expressed as a linear error at the 90% confidence level. This value is what is input into the header as the relative vertical accuracy.

6.4.2 Attribution. Name of attribute category required by the feature as specified in the Feature Attribution Coding Standard (FACS). Attribute categories are characteristics in menu form relative to a specific feature or features.

6.4.3 Attribute Code (Acode). Three digit alpha or alphanumeric character (acronym) FACS code assigned to each attribute category which identifies the attribution category (e.g., ZVL-Z Value). Attribute categories are defined by mutually exclusive sets of attribute values which are feature dependent.

6.4.4 Cell. 1° by 1° area of coverage.

6.4.5 Circular error. An accuracy figure representing the stated percentage of probability that any point expressed as a function of two linear components (for example, latitude and longitude or northing and easting) will be within the given figure.

6.4.6 DTED. A geographic matrix of terrain elevation data points converted into a numerical format for computer storage and analysis at precise increments of latitude and longitude. The precise increment depends upon the latitude. Elevation values are expressed in meters.

a. DTED Level 1 - Precise 3 arc second horizontal position referenced to the origin of the southwest corner, evenly spaced in latitude and longitude in a South to North profile sequence.

b. DTED Level 2 - Precise 1 arc second horizontal position referenced to the origin of the southwest corner, evenly spaced in latitude and longitude in a South to North profile sequence.

6.4.7 Inclusion condition. Conditions under which the feature/ attribute(s) are required by the product.

a. FACS Feature code: 9B045

(1) Category: General (9)

(2) Sub category: Control points (B)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.

b. Feature: Elevation Point

c. Feature Definition: A point within a matrix of points having a precise position on the earth's surface with the elevation relative to MSL.

d. Feature: Point

e. Required Attribute: ZVL (Z Value). The elevation of the feature as referenced to MSL.

(1) Increment: 1 meter

(2) Limits: From Mean Sea Level to the tallest portion of the feature.

(3) Variance: NA

(4) Range: -400 . . . 9998

6.4.8 Linear error. Linear error is the difference between the true or known value and the measured or derived value, and is normally expressed in terms of a percentage probability level. For example, LE 90% is the term used to express the linear error at 90% probability, which is the Map Accuracy Standard. This refers to the vertical accuracy of terrain elevations in the digital data base.

6.4.9 Mean Sea Level (MSL). The average height of the surface of the sea for all stages of the tide, used as a reference for elevations.

6.4.10 Nautical mile. 6076 feet; 1852 meters (international value).

6.4.11 Off-line digital data base. Magnetic media containing information in digital form including header information, geographic coordinates and descriptive information for planimetric features and terrain within a specific geographic area.

6.4.12 WGS (World Geodetic System). A consistent set of parameters describing the size and shape of the Earth, the positions of a network of points with respect to the center of mass of the Earth, transformations from major geodetic datums, and the potential of the Earth (usually in terms of harmonic coefficients).

6.5 Subject term (key word) listing.

CD-ROM
1 degree by 1 degree (1° by 1° cell) cell

6.6 Standardization agreements. Certain provisions of this specification may be subject to international standardization agreement. When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

6.6.1 International Standardization Agreements (STANAGs).

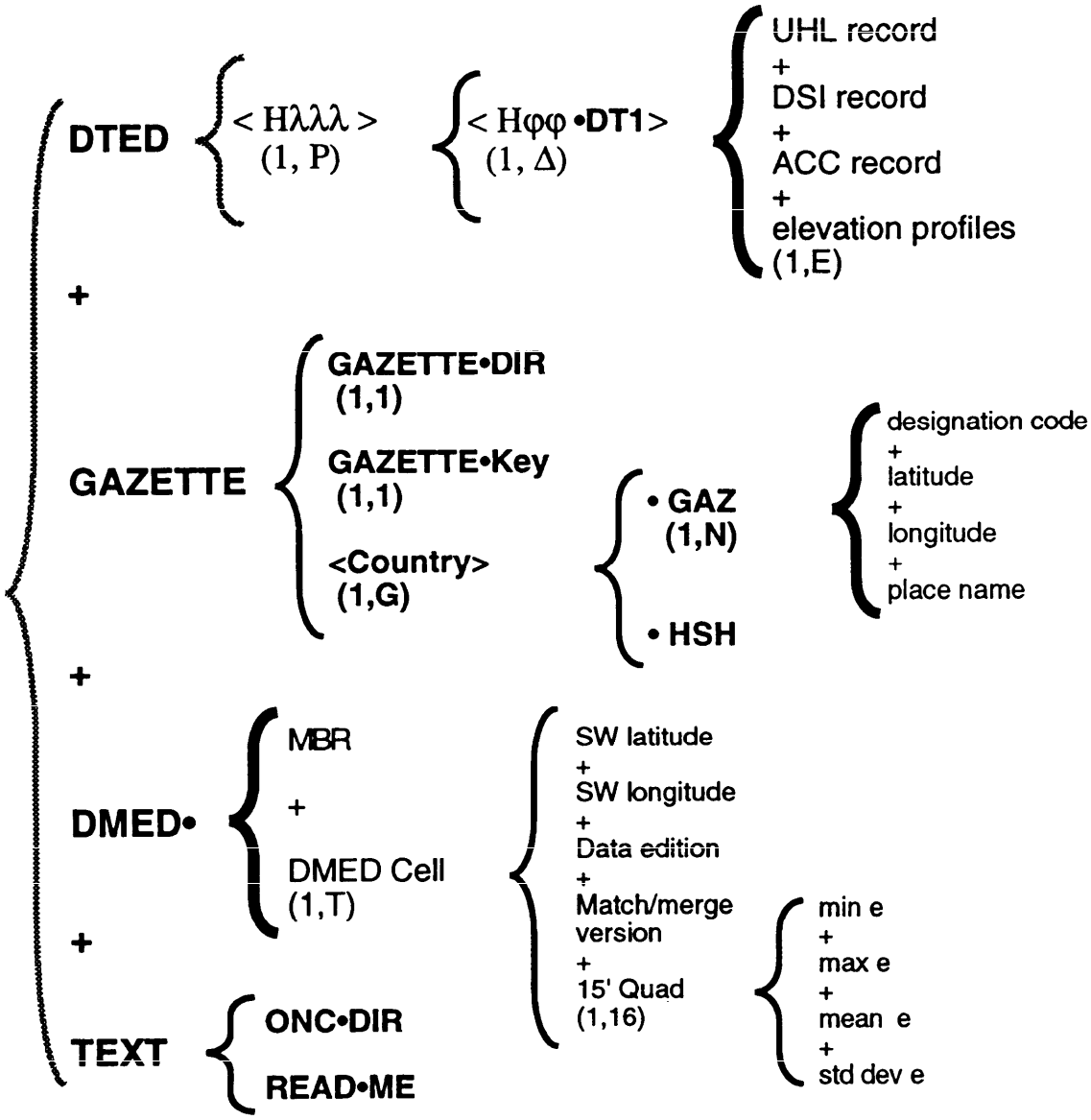
a. DTED Level 1 and DTED Level 2 -

(1) STANAG 3809, "Digital Terrain Elevation Data Exchange Format".

(2) STANAG 2211, "Geodetic Datums, Spheroids, Grids and Grid References".

6.7 Changes from previous issue. The margins of this specification are marked with side bars to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever from any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:54Z
Check the source to verify that this is the current version before use.



Hλλλ- hemisphere (E or W) and longitude
Hφφ- hemisphere (N or S) and latitude
P- number of 1° cell columns within the disc, a multiple of 6
Δ- number of 1° DTED cells within the columns, a multiple of 6
E - number of profiles per cell (latitude zone dependent, 1201 elevations per profile)
G - number of country/region gazettes on the disc
N - number of place names within the country/region gazette
T - total number of 1° cells on disc = P x Δ, a multiple of 6

FIGURE 1. DTED CD-ROM logical data file structure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

INDEX

	PARAGRAPH	PAGE
Absolute accuracy	3.3.1	3
Absolute horizontal accuracy	3.3.1.1	3
Absolute vertical accuracy	3.3.1.2	3
Accuracy	3.3	3
Accuracy	6.4.1	32
Accuracy description record (ACC)	3.13.5	25
Accuracy sub region description	3.13.5.1	25
Acquisition Requirements	6.2	32
Adjacent data files	3.6.1	4
APPLICABLE DOCUMENTS	2.	1
Attribution	6.4.2	33
Attribute Code (Acode)	6.4.3	33
CD-ROM characteristics	3.14.1.2	26
CD labeling	3.14.3.1	27
CD-ROM content	3.14.1.1	26
CD-ROM labeling	3.17	31
CD-ROM labeling and packaging	3.14.3	27
Changes from previous issue	6.7	35
Cell	6.4.4	33
Circular error	6.4.5	33
Classification	1.3	1
Classification of inspection	4.1	31
Conformance inspection	4.3	31
Copyright	3.16	31
Data block checksum	3.10.6	7
Data file structure	3.10.1	7
Data record characteristics	3.11	7
Data record sequence	3.10.5	7
Data records	3.8.2	5
Data Set Identification (DSI) Record	3.13.4	23
Data value sequence	3.10.4	7
Datum	3.2	3
Definitions	6.4	32
Description of record contents	3.12	11
Digital Mean Elevation Data (DMED)	3.14.5	28
Directories	3.14.2	27
DMA Customer Help Desk	3.5	4
DMED file	3.15.4.1	28
DTED	6.4.6	33
DTED files	3.14.4	27
DTED on Compact Disc-Read Only Memory (CD-ROM)	3.14	26
Eight Millimeter (8 mm) Tape	3.7.1.2	5
Elevation values	3.11.2	8
Examination	4.3.1	31
Explanation of records and fields (DTED)	3.13	23
Fields(DSI)	3.13.4.1	23
Fields(UHL)	3.13.3.1	23
File description	3.9	6
File header label (HDR)	3.13.2	23
File structure diagram	3.10.7	7
Files extent	3.8.2	7
First article	3.1	3
First article inspection	4.2	31
Gazette directory	3.15.6.1	29

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

INDEX

	PARAGRAPH	PAGE
GAZETTE.DIR	3.15.6.5	30
GAZETTE.KEY	3.15.6.4	30
Gazetteer	3.15.6	29
General	2.1	1
General	3.14.1	26
Government documents	2.2	1
Government furnished material	4.4	32
Government property surplus	4.5	32
Hash file	3.16.6.3	29
Horizontal datum	3.2.1	3
Inclusion condition	6.4.7	33
Individual cell record	3.15.4.2	28
Information booklet	3.17.1	31
Intended use	6.1	32
International Standardization Agreements (STANAGs)	6.6.1	35
Linear error	6.4.8	34
Logical characteristics	3.10	7
Marking	5.2	32
Mean Sea Level (MSL)	6.4.9	34
Missing Elevations	3.11.3.2	8
Nautical mile	6.4.10	34
Nine track(9) tape	3.7.1.1	4
Non-Government publications	2.3	2
NOTES	6.	32
Null elevations	3.11.3.1	8
Numeric value	3.11.1	8
Off-line digital data base	6.4.11	34
ONC.DIR	3.15.7.2	30
Order of precedence	2.4	3
Other Government documents, drawings, and publications	2.2.2	2
PACKAGING	5.	32
Packaging	5.1	31
Partial cells	3.11.3	8
Physical characteristics of magnetic tape	3.7.1	4
Primary file	3.16.6.2	29
Product format	3.6	4
Purpose	1.2	1
READ.ME	3.15.7.1	30
Record characteristics on Magnetic Tape	3.8	5
Recorded labels	3.8.1	5
Relative accuracy (point-to-point)	3.3.2	3
Relative horizontal accuracy (point-to-point)	3.3.2.1	4
Relative vertical accuracy (point-to-point)	3.3.2.2	4
REQUIREMENTS	3.	3
SCOPE	1.	1
Scope	1.1	1
Security	3.4	4
Security classification	3.4.1	4
Specifications, standards, and handbooks	2.2.1	2
Standard format	3.7	4
Standardization agreements	6.6	34
Subject term (key word)listing	6.5	34
Supersession	6.3	32

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

INDEX

	PARAGRAPH	PAGE
Terrain elevation intervals	3.10.3	7
Tests	4.3.2	32
Textual information	3.15.7	30
Type of Input/Output (I/O)	3.7.2	5
User Header Label (UHL)	3.13.3	23
User's responsibilities	3.15.8	30
VERIFICATION	4.	31
Vertical datum	3.2.2	3
Volume header label (VOL)	3.13.1	23
WGS (World Geodetic System)	6.4.12	34

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

Concluding Material

Custodian:
DMA - MP
Army - TI
Air Force - 09
Navy - NO

Review Activities:
Marine Corps - MC

Preparing Activity
DMA - MP

(Project MCGT-0161)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-89020A	2. DOCUMENT DATE *(YYMMDD)* 960418

3. DOCUMENT TITLE Performance Specification for Digital Terrain Elevation Data (DTED)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(if applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME Defense Mapping Agency ATTN: ATIS, MS A-10	b. TELEPHONE *(Include Area Code)* (1) Commercial (703) 275-8509	(2) AUTOVON 235-8509
c. ADDRESS *(Include Zip Code)* 8613 Lee Highway Fairfax, VA 22031-2137	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA. 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198-290

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:54Z
Check the source to verify that this is the current version before use.